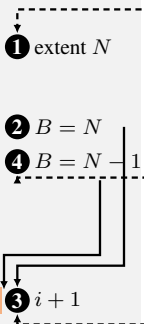


(a) Patch and array extent

```
M1 // N := MAX_PIPES * 2
M2 struct dc_link *links[N];
M3 // get_host_router_...(...)
M4 ...
M5 - for (i=0; i<N; ++i) {
M6 + for (i=0; i<N-1; ++i) {
M7 if (!dc->links[i] || ...)
M8 continue; ...
M9 if (hr_index_temp == ...)
M10 primary=dc->links[i];
M11 secondary=dc->links[i+1];
M12 ...;
M13 break;
M14 }
M15 }
```

Vulnerable: $\max(i+1) = N$

Fixed: $\max(i+1) = N - 1$



(b) Starting checker

```
B1 checkASTCodeBody(F) :
B2 for loop i < B:
B3 find access A[i+1]
B4 if no body guard:
B5 report(A[i+1])
```

5 Missing extent test; V/F: 1/1

(c) Automatic refinement

```
C1 checkASTCodeBody(F) :
C2 keep loop/access discovery
C3  $N = \text{type\_extent}(A)$ 
C4  $\text{max\_index} = (B-1)+1$ 
C5 if  $\text{max\_index} < N$ :
C6 suppress; else: warn
```

Tested guards normalized; no name filter

6 Extent-aware condition; V/F: 1/0